

Jay Niketan Pathare

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EXPERIENCE

Graduate Student Developer | HeinOnline (CSE 611 Project), Buffalo, NY

January 2026 – April 2026

- Engineered a fault-tolerant OCR/LLM metadata extraction pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC data from law journals (30k+ pages).
- Merged outputs from all three sources into one validated structure with fuzzy matching, cutting manual data-entry work by ~90% and pushing extraction accuracy to 91%+ over five iterations, benchmarked against ground truth.
- Parallelized the pipeline with a worker pool that pulls the next journal from a shared queue on completion and added an “aiscore” system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Led delivery of platform modules end to end: gathered requirements in client meetings, prototyped to validate scope, and broke the finalized work down across a small team.
- Reviewed and merged team pull requests alongside my own code, catching issues and maintaining code quality.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records on normalized relational schemas for fast retrieval and reporting, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
- Delivered a real-time telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation, and added a color-blindness diagnostic module (97%+ accuracy) with automated reporting for practitioners.
- Containerized 3+ production projects using Docker and Docker Compose, keeping local, staging, and deployment environments consistent and reducing new-developer setup time by ~90%.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups for secure, low-latency communication between backend services.
- Automated cloud resource provisioning, cutting setup time by ~40% and reducing deployment issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, producing reports on key operational trends.

ACADEMIC PROJECTS

LLM Application Optimization Engine

September 2026

- Built a multi-model LLM pipeline that turns a job description into a tailored resume and cover letter, using dynamic programming to spend a fixed budget on requirements not yet proven elsewhere.
- Cut end-to-end runtime by ~73% (~11 to ~3 minutes) using three concurrent LLM workers with account failover, regenerating only failed output and holding each run to 5 base model calls.
- Wrote Python checks that reject any generated claim or metric not backed by source evidence before it ships, covered by a 2,500+ test pytest suite with metamorphic tests and no live API calls.

LaunchOps: Office Onboarding & Real-Time Learning Platform

July 2026

- Developed a Django REST and React onboarding platform running a 20-task workflow across four roles, with server-side checks that stop users from completing another role’s tasks or skipping prerequisites.
- Added a Go WebSocket live-quiz service running two replicas behind nginx, coordinated through Redis Pub/Sub and Lua deadline checks, with a unique MongoDB index rejecting duplicate answers.

Fraud Detection using Graph Neural Networks

March 2026

- Trained a GraphSAGE fraud model with Focal Loss for a ~27.6:1 class imbalance, reaching 0.9259 AUC-ROC against ~0.858 for an MLP baseline, while GAT overfit and collapsed around hub nodes.
- Turned flat transaction rows into a graph by linking records that share a card, address or email domain, capping oversized groups to keep memory in check across 590k+ IEEE-CIS transactions.

EDUCATION

Master’s in Computer Science

August 2025 – February 2027

University at Buffalo, The State University of New York, Buffalo, NY

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society’s Institute of Technology, Mumbai University, India

August 2019 – May 2023

TECHNICAL SKILLS

- Languages & Databases:** Python, Go, JavaScript, C, SQL, PostgreSQL, MongoDB, MySQL
- Frameworks & Libraries:** Django, Django REST Framework, Django Channels, ReactJS, PyTorch, scikit-learn, Pandas
- AI/ML:** LLM Integration, Agentic AI, Prompt Engineering, OCR (Tesseract), Machine Learning, Deep Learning, GNNs (GraphSAGE, GAT), Model Evaluation
- Software Engineering:** Distributed Systems, System Design, REST APIs, Concurrency, Parallel Processing, OOP, Unit Testing (pytest), Load Testing, ETL, Agile, Real-Time Monitoring
- Tools & Cloud:** AWS (EC2/RDS), Docker, Docker Compose, Redis, Linux, Git, GitHub, WebSockets, WebRTC, JWT, Gemini API, Groq API

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Certified Cloud Practitioner** July 2024
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Apr. 2023 (97.68% accuracy). [Link](#).